

443/1

AGRICULTURE

PAPER 1

(Theory)

TIME: - 2 hours

MARCH 2026

MARKING SCHEME

SECTION A 30MARKS (Answer all the questions in this section in the space provided.)

1. State **four** disadvantages of intensive farming system. (2 marks)
 - If management is poor, it can lead to big losses.
 - .It requires high initial capital.
 - It requires high labour costs/labor expensive.
 - .Requires high level of management /skilled labor. $4 \times \frac{1}{2}$
2. State four advantages of nomadic pastoralism. (2 marks)
 - Requires little capital.
 - .Farmers can keep as many animals as possible.
 - .Animal waste is evenly distributed all over the field to allow grass regeneration.
 - .There is no disease and pest build up/increase.
 - .Pasture is given time to regenerate
3. State four characteristics of extensive farming system. (2 marks)
 - Large tracts/ pieces of land.
 - Low capital investment.
 - Low labour per unit area. $4 \times \frac{1}{2}$
 - Low yield per unit area.
4. State two causes of hard pans in the soil. (1 mark)
 - Shallow tillage/ cultivation at the same level continuously.
 - Continuous use of heavy machines on wet soil. $1 \times \frac{1}{2}$
5. State 2 reasons why green manure is not commonly used. (1 mark)
 - Most of the green manure crops are food crops. $1 \times \frac{1}{2}$
 - Green manure crops may use available nutrients and water in the soil and leave very little for the next crop.
 - They release nutrients into the soil slowly.
 - Most nutrients are used by micro organisms in decomposing green manure.
 - It takes time for green manure crop to decompose hence planting is delayed.

6. State the two types of inventory records. (1 mark)
- Consumable goods inventory records
 - Permanent goods inventory records
7. Give four reasons for growing crops under optimum temperature. (2 marks)
- Enhances seed germination /emergence.
 - Promotes soil microbial activities.
 - Enhances vigorous growth and development.
 - Enhances high yields.
 - Minimizes incidences of certain pests and diseases.
8. State four ways through which HIV/AIDS affect agriculture negatively. (2 marks)
- Shortage of labour.
- Lack of motivation to invest in agriculture.
 - Increased cost of living leading to low investment in agriculture/ lack of resources for agricultural production.
 - The government and NGOs spend a lot of time and resources to control the disease instead of investment in agriculture.
 - Lack of market for agricultural produce.
 - Less time spent on farming activities as people cater for the sick.
9. List two aspects of light that affect agriculture. (1 mark)
- Light intensity
 - Light wavelength/light frequency
 - Light duration/photoperiodism
10. State two methods of water conveyance. (1 mark)
- Use of pipes/ piping.
 - Use of containers.
 - Use of canals.
11. State four advantages of drip/trickle irrigation. (2 marks)
- Little amount of water is required (compared to other types of irrigation).
 - Water under low pressure can be used (so long as it flows along the pipes).
 - It discourages fungal diseases (e.g. coffee berry disease) because it does not wet the leaves.
 - It discourages the growth of weeds between the rows.
12. State four deficiency Symptoms of nitrogen. (2 marks)
- Leaf chlorosis- Yellowing of the leaves
 - Stunted growth – plants become dwarfed
 - Premature ripening of fruits
 - Premature shedding of the leaves.
 - Production of light seeds in cereals
 - Production of anthocyanin leading to purpling of leaves.

13. List four examples of nitrogenous fertilisers. (2 marks)

- Sulphate of Ammonia (Sulphate of Ammonia)
- Ammonium Sulphate Nitrate (ASN)
- Calcium Ammonium Nitrate (CAN)
- Urea

14. State two vegetative materials used to propagate pineapple. (1 Mark)

- Crown
- slip
- Sucker

1 × ½

15. State **four** methods of breaking seed dormancy. (2 marks)

- Soaking in water
- Chemical treatment
- Heat treatment
- Mechanical method/scarification

4 × ½

16. Distinguish between a seed bed and a seedling bed. (1 mark)

Seed bed- This is a piece of land (large or small) which has prepared to receive the planting materials that are allowed to grow until harvesting

Seedling bed- This is a special type of nursery bed used for raising seedlings that have been removed from the nursery bed due to overcrowding before they are ready for transplanting.

17. List four characteristics of a good site for nursery bed. (2 marks)

- Near the water source.
- Deep, well drained and fertile soil.
- Gentle slope/ topography.
- Secure against animals and birds.
- Area where crops of the same family have not been grown for 3 years.
- Sheltered against strong wind and sun.
- Easily accessible.

4 × ½

18. List any two pest that attack tomatoes. (1 mark)

- Nematodes-
- Cutworm-
- Redspiderandmites
- Cutworm
- Americanbollworm

1 × ½

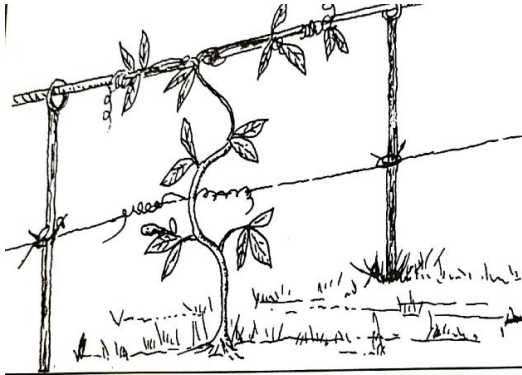
19. List any four field practices carried out on carrots. (2 marks)

- Pest control-
- Irrigation-
- Top-dessing-
- Earthingup-
- Weeding –
- Thinning-

4 × ½ mks

SECTION B 20 MARKS (Answer all the questions in this section in the space provided)

20. The diagram below shows a field practice carried out on certain crops. Use it to answer the questions that follow.



a) Identify the field practice.

- Trellising

(1 mark)

1×1

b) Name any two crops where this practice can be done.

(2 marks)

- Passion plants.
- Grape vines.
- Cucumbers.
- Some bean varieties.
- Pumpkins.
- Gourds.

2×1

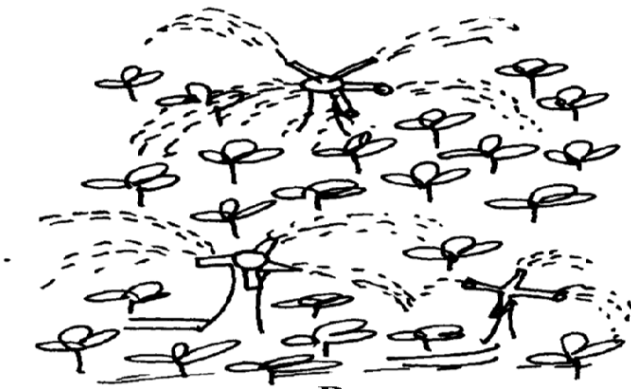
c) Name two other field practices used to train a crop.

(2 marks)

- Propping
- Staking

2×1 mks

21. Below is a diagram showing a method of irrigation.



a) Identify the method of irrigation shown above.

(1 mark)

- Overhead/ sprinkler method

1×1

b) State **four** disadvantages of the above system compared to furrow irrigation.

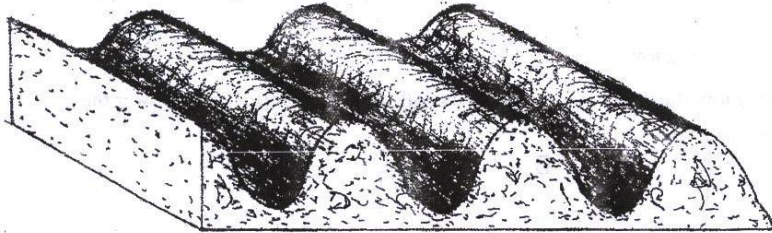
(4 marks)

- It is expensive to install e.g. buying pipes, sprinklers, and water pump.

- It encourages fungal diseases because water comes into contact with leaves.
- It can cause soil erosion in sloppy areas if not well controlled.
- It requires wind breakers to prevent water drift.
- Requires skill to maintain.

4×1 mks

22. The diagram below shows a tertiary operation carried out during land preparation.



a) Identify the tertiary operation shown above.

(1 mark)

- Ridging

1×1

b) Describe how the above operation is carried out.

(2 marks)

- Soil is dug in a continuous line forming a furrow.
- Soil is heaped on one side forming a heap/ bund.

2×1 mks

c) Give two reasons why the above operation is carried out. (2 marks)

- It encourages root development and expansion of tubers.
- It makes harvesting of root crops easy.
- It helps to conserve soil and water.
- It prevents soil erosion.
- It helps to improve soil drainage.
- It prevents greening of tubers

2×1 mks

23. Below is a diagram showing a certain soil structure.



a) Identify the soil structure shown above.

(1 mark)

Platy soil structure.

1×1

b) State two disadvantages of the soil structure shown above. (2 marks)

- It hinders root penetration,
- It hinders drainage
- It hinders aeration.

2×1 mks

c) Name any two other types of soil structure.

(2 marks)

- Single grained structure.
- Crumbly structure.
- Granular structure.
- Prismatic structure.
- Columnar structure.
- Blocky structure

2×1 mks

SECTION C 40 MARKS (Answer any two questions in this section in the space provided after question 28)

24. a). Describe how the following climatic factors influence agriculture.

i) Temperature

(4 marks)

Growing crops under optimum temperature:-

- Enhances seed germination /emergence.
- Promotes soil microbial activities.
- Enhances vigorous growth and development.
- Enhances high yields.
- Minimizes incidences of certain pests and diseases.

4×1 mks

ii) Light.

(6 marks)

- Light intensity- refers to the strength of light.
 - Increase in light intensity increases the rate of photosynthesis hence faster growth of crops.
 - Inadequate or lack of light cause etiolation (thin and pale crops)
- Light duration/photoperiodism- refers to the period during which light is available to the crops per day.
 - It determines the type of crop to grow i.e. short day, long day or neutral day plants.
- Light wavelength/light frequency- refers to the quality of light e.g. ultra violet (UV) and infra red wavelengths.
 - Chlorophyll absorbs certain wavelengths of light.
 - If those wavelengths are not available

6×1 mks

b). Explain five properties of nitrogenous fertilisers.

(5 marks)

- They are highly soluble in water.
- They are easily leached hence it is applied as a top dress.
- Has scorching (burning) effect on young crops during wet seasons.
- They are highly volatile especially during hot season. That is, they change into gaseous forms.
- They have a short residual effect hence applied frequently.
- They are hygroscopic, that is, they have a tendency to cake under moist conditions hence should be stored in dry conditions.
- They are highly corrosive since they can corrode the skin and any other metallic

surfaces. Therefore, they should not be handled with bare hands.

5×1mks

c). State five pieces of information contained in the field operation records. (5 marks)

- Area of the field
- Crops planted.
- Date of ploughing.
- Date of planting.
- Fertilizers used, amount applied and date of application.
- Type of seed used, the amount and cost.
- Diseases and pests identified and cost of control.
- Weeding dates, methods, and cost.
- Harvesting date, quantities of produce.
- Labour. How many workers are hired and their wages.

5×1 mks

25. a) Explain four disadvantages of using organic manures. (4 marks)

- Bulkiness- they occupy large storage space which makes it difficult to store, transport and apply.
- Laborious in transport and application- they require a lot of labor to transport and apply increasing the cost of production.
- They spread pests, diseases and weeds- if materials used to make manures are infested with pests, diseases and weed seeds they spread them to the farm.
- Loss of nutrients- they can lose nutrients easily through leaching and volatilization if manures are poorly stored.
- They cannot benefit crops if not fully decomposed- because they release nutrients that can scorch the crop.
- Have low nutrient per unit volume- this means that they require a lot of manure to supply the required amount of nutrients.
- Difficult to quantify/calculate amount of nutrients per unit volume- because the amount of nutrients contained varies.

4×1

b). Explain six factors that influence crop rotation. (6 marks)

- Crop nutrient requirement-Heavy/gross feeders should come early in the rotation program.
- Pest and disease control-Crops of the same families should not follow each other in the rotation program because they are affected by the same pests and diseases.
- Soil fertility- Legumes should be included in the rotation program to fix atmospheric nitrogen hence improve fertility.
- Crop root depth- Deep rooted crops should be rotated with shallow rooted crops (to ensure maximum utilization of nutrients).
- Weed control- Cover crops should be included in the rotation program to smother weeds. Crops with associated weeds should not follow each other in the program.
- Soil structure- Grass ley should be included in the rotation program to hold soil particles together hence improving soil structure.

6×1mks

c). Describe five field management practices carried out on onions.

(10 marks)

- Thinning-Carry out thinning if planting is done directly to reduce competition.
- Weeding-Carry out weeding through shallow cultivation taking care not to damage the shallow roots. Remove excess soil around the roots region to facilitate bulb expansion. Do not compact the soil around the bulb.
- Top-dressing-Top-dress using CAN at appropriate rates. It is done 3 months after planting.
- Watering-Carry out watering during dry season.
- Pest control-onion thrips(Thripstabaci) cause silvering and withering of leaves from tips downwards. Control by spraying with appropriate insecticides e.g. Diazinon or Fenithion.
- Disease control. Purple blotch is characterized by oval greyish lesions with purple centers on leaves causing leaf curling and die back. Control by crop rotation and use appropriate fungicides. Downy mildew is characterized by brown spores covering the leaves leading to death of a whole plant. Control by crop rotation and use appropriate fungicides.

5×2

26. a) Explain ten nursery management practices. (10 marks)

- Hardening off- This is a practice of preparing seedlings to adapt to the ecological conditions prevailing in the seed bed.
It is done by gradual reduction of shade and watering 1-2 weeks before transplanting.
- Shading- a light shade is erected/ constructed over the nursery bed to protect young seedlings from excess sunlight/ rain drops that damage them
Dark conditions should be avoided to prevent etiolation.
- Pricking out- Refers to the practice of removing overcrowded seedlings from a nursery bed and planted in a second nursery bed/ seedling bed to avoid competition.
- Mulching- A light mulch is applied on the nursery bed before seedlings emerge to conserve moisture and control soil erosion.
- Removal of mulch- mulch is removed immediately the seedlings start to emerge to prevent etiolation.
- Weed control- weeds are removed through uprooting to minimize competition with the crop.
- Watering- a nursery bed is watered regularly in the morning and evening to ensure adequate water supply.
- Pest control- controlled through application of appropriate chemicals/ fungicides to keep the seedlings healthy.
- Disease control- controlled through application of appropriate chemicals/ fungicides to keep the seedlings healthy.
- Root trimming- cut tap roots to encourage development of lateral branching of roots and ease lifting of seedlings.
- Application of foliar feed fertilizer- done to boost growth and development of the seedling.

10×1

b). Explain five factors that influence seed rate. (5 marks)

- Seed purity- when planting pure seeds, less seed rate is required.
- Germination percentage- less seed rate is required when germination percentage is higher. .
- Method of planting- broadcasting method require higher seed rate than row planting
- Spacing- Closer spacing requires higher seed rate than wider spacing.
- Number of seeds per hole- Two or more seeds per hole require a higher seed rate than one seed per hole.
- Purpose/intended use of the crop- Crops used as fodder are closely spaced hence require higher seed rate.

5×1mks

c.) Explain **five** advantages of using vegetative materials for crop propagation. (5 marks)

- Crops originating from vegetative materials mature faster than those from seeds.
- The use of vegetative materials is easier and faster, especially where seeds show prolonged dormancy.
- The crops show uniformity in qualities e.g. disease resistance, seed size, color, keeping/ storage quality and chemical composition
- It is possible to produce many varieties of compatible crops on the same roots stock.
- Facilitates the propagation of crops that are seedless or those that produce that are not viable or have long dormancy period e.g. sugarcane, bananas, napier grass.
- The resulting plant has desirable shape and size for ease of harvesting and spraying.

5×1mks